Retrieval-Augmented Generation: An Overview

What RAG is

Retrieval-Augmented Generation (RAG) is a technique in which a language model answers a question using passages retrieved from an external knowledge base instead of relying only on what it memorised during training. The term was introduced by Lewis and colleagues in 2020, who combined a sequence-to-sequence generator (parametric memory) with a dense vector index of Wikipedia (non-parametric memory). The central idea is that knowledge can be updated by changing the index, without retraining the model.

The standard pipeline

A typical RAG system has two phases. In the offline indexing phase, documents are loaded, cleaned, split into chunks, converted into embedding vectors and stored in a vector index. In the online query phase, the user question is embedded with the same model, the nearest chunks are retrieved by similarity, the chunks are assembled into a prompt together with the question, and the language model generates an answer that cites the retrieved sources.

Why use RAG

RAG has three practical benefits. First, answers can be traced back to named source documents, which makes them auditable. Second, the knowledge base can be extended or corrected by re-indexing documents, which is far cheaper than fine-tuning. Third, grounding the model in retrieved text reduces, but does not eliminate, hallucination.

Typical failure modes

RAG systems fail in three characteristic ways. A retrieval miss happens when the relevant passage is not among the top-k results, so the generator never sees the evidence. Context overflow happens when too many or too long passages exceed the model context window and are truncated. Unfaithful generation happens when the model states something that the retrieved passages do not support. Each failure should be measured separately, because a good final answer can hide a weak retriever and a bad answer can come from a strong retriever.

Handling questions without an answer

A well-behaved RAG system must be able to abstain. When the best retrieved passages have low similarity to the question, or when the passages do not contain the requested fact, the assistant should say that the knowledge base does not contain the answer instead of guessing. Abstention is typically implemented with a minimum similarity threshold on retrieval combined with an instruction in the prompt to refuse when the context is insufficient.

Scope of this study guide

This study guide is a small synthetic corpus written for a mini project. It explains chunking, embeddings, vector search, retrieval metrics, prompting and security in separate documents so that a question can be answered from one document or require combining several.